

SWAPNONEEL MONDAL

Computer Science Scholar • High-Performance Web & Full-Stack Developer • AI Systems

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PROFESSIONAL SUMMARY

Computer Science student at Institute of Engineering & Management (IEM), Kolkata, focusing on modern component-driven frontend architecture, cognitive AI integrations, digital systems, and full-stack web platforms. Experienced in high-velocity prototyping (“vibe coding”), combining core algorithmic foundations with LLMs (Gemini, Claude), Web Audio API synthesizers, and real-time computer vision models to engineer ultra-responsive, accessible digital applications.

EDUCATION

Bachelor of Technology in Computer Science & Engineering — Institute of Engineering & Management (IEM), Kolkata **2024 – Present**

- **Foundational Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Digital Logic Design, Computer Architecture, Discrete Mathematics.
- **Practical Engineering:** Algorithm implementations in C & Python, matrix analysis in MATLAB, digital gate & circuit simulation in Logisim.

FEATURED PROJECTS

Scholar OS — Cognitive Academic OS & Vision Analyzer [Next.js 16, React 19, Gemini 2.5 Flash, TypeScript, Tailwind CSS] **2025 – 2026**

Live Demo: scholardashboard.vercel.app • Repository: github.com/KaiX-Jr/scholar-os

- Architected an AI-native academic operating system that converts blackboard/whiteboard photos into organized markdown notes, summaries, and key formula sheets via Gemini multimodal vision.
- Implemented attendance threshold forecast simulations, subject analytics tracker, and daily automated AI quiz generation with real-time scoring.

Dynovision — Real-Time Computer Vision & Gesture AI HUD [React 19, MobileNetV2, MediaPipe, HTML5 Canvas, Netlify] **2025 – 2026**

Live Demo: dynovision.netlify.app • Platform: Web-based Client ML Engine

- Built an in-browser affective computer vision engine executing facial landmark detection, emotion classification, and gesture tracking directly on client hardware with zero backend latency.
- Designed a telemetry HUD with dynamic canvas overlays, real-time FPS monitoring, and glassmorphic controls.

LogiSimulate — Digital Logic Circuit Simulator & Minimizer [Python, C, Logisim, Digital Systems] **2024 – 2025**

- Developed a circuit simulator for testing combinational boolean networks, multiplexers, and logic gate propagation delays.
- Implemented automated Karnaugh map simplification algorithms to minimize gate counts and optimize hardware efficiency.

DataPredict ML — Predictive Modeling & Matrix Analytics [Python, MATLAB, NumPy, Scikit-learn] **2024 – 2025**

- Engineered data pipelines for statistical exploratory data analysis, matrix decomposition, and multi-variable predictive regression models.

EXPERIENCE & TECHNICAL ENDEAVORS

Vibe Coder & AI-Assisted Full-Stack Developer — Independent & Open Source Projects **2026 – Present**

- Leveraging state-of-the-art LLMs and agentic workflows to rapidly architect, prototype, and ship full-stack web applications with high test coverage and component modularity.
- Synthesized Web Audio API spatial synthesizers, spring physics animations (Framer Motion), and responsive Bento grid interfaces with zero-runtime CSS variables.

Independent Web Creator & Frontend Developer — Self-Directed Engineering **2024 – Present**

- Designing and deploying modern web applications adhering to Core Web Vitals (CWV), strict accessibility standards (a11y), and cross-browser responsiveness across desktop and touchscreens.

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), TypeScript, Python, C, MATLAB, HTML5, CSS3/PostCSS
Web & Frameworks: React 19, Next.js 16 (App Router), Tailwind CSS v4, Framer Motion, Anime.js, Web Audio API
AI & Computer Vision: Google Gemini 2.5 Flash API, Prompt Engineering, MobileNetV2, MediaPipe, LLM Integration
Tools & Environments: Git, GitHub, Turbopack, Logisim, Netlify, Vercel, VS Code, Windows/PowerShell